FAIRCHILD SEMICONDUCTOR™

March 1996

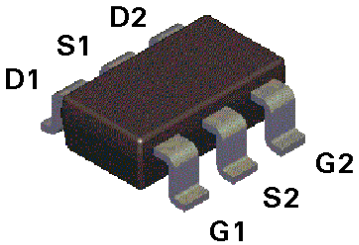
# NDC7003P
# Dual P-Channel Enhancement Mode Field Effect Transistor

## General Description

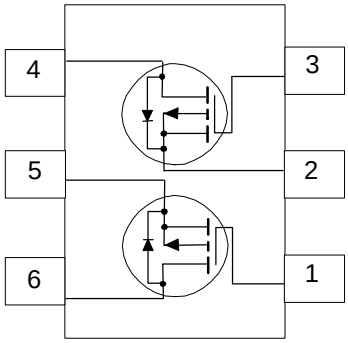
These dual P-Channel enhancement mode power field effect transistors are produced using Fairchild's proprietary, high cell density, DMOS technology. This very high density process has been designed to minimize on-state resistance, provide rugged and reliable performance and fast switching. This product is particularly suited to low voltage applications requiring a low current high side switch.

## Features

- -0.34A, -50V. $R_{DS(ON)}$= 5Ω @ $V_{GS}$=-10V.
- High density cell design for low $R_{DS(ON)}$.
- Proprietary SuperSOT™-6 package design using copper lead frame for superior thermal and electrical capabilities.
- High saturation current.




SOT-6 (SuperSOT™-6)




**Absolute Maximum Ratings** $T_A$ = 25°C unless otherwise noted

| Symbol | Parameter | | NDC7003P | Units |
|---|---|---|---|---|
| $V_{DSS}$ | Drain-Source Voltage | | -50 | V |
| $V_{GSS}$ | Gate-Source Voltage - Continuous | | -20 | V |
| $I_D$ | Drain Current - Continuous | (Note 1a) | -0.34 | A |
| | - Pulsed | | -1 | |
| $P_D$ | Maximum Power Dissipation | (Note 1a) | 0.96 | W |
| | | (Note 1b) | 0.9 | |
| | | (Note 1c) | 0.7 | |
| $T_J,T_{STG}$ | Operating and Storage Temperature Range | | -55 to 150 | °C |
| **THERMAL CHARACTERISTICS** | | | | |
| $R_{\theta JA}$ | Thermal Resistance, Junction-to-Ambient | (Note 1a) | 130 | °C/W |
| $R_{\theta JC}$ | Thermal Resistance, Junction-to-Case | (Note 1) | 60 | °C/W |

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**ELECTRICAL CHARACTERISTICS** ($T_A$ = 25°C unless otherwise noted)

| Symbol | Parameter | Conditions | Min | Typ | Max | Units |
|---|---|---|---|---|---|---|
| **OFF CHARACTERISTICS** | | | | | | |
| $BV_{DSS}$ | Drain-Source Breakdown Voltage | $V_{GS}$ = 0 V, $I_D$ = -250 µA | -50 | | | V |
| $I_{DSS}$ | Zero Gate Voltage Drain Current | $V_{DS}$ = -40 V, $V_{GS}$ = 0 V | | | -1 | µA |
| | | $T_J$ = 125°C | | | -500 | |
| $I_{GSSF}$ | Gate - Body Leakage, Forward | $V_{GS}$ = 20 V, $V_{DS}$ = 0 V | | | 100 | nA |
| $I_{GSSR}$ | Gate - Body Leakage, Reverse | $V_{GS}$ = -20 V, $V_{DS}$ = 0 V | | | -100 | nA |
| **ON CHARACTERISTICS** (Note 2) | | | | | | |
| $V_{GS(th)}$ | Gate Threshold Voltage | $V_{DS}$ = $V_{GS}$, $I_D$ = -250 µ.A | -1 | -2.5 | -3.5 | V |
| | | $T_J$ = 125°C | -0.8 | -2.2 | -3 | |
| $R_{DS(ON)}$ | Static Drain-Source On-Resistance | $V_{GS}$ = -10 V, $I_D$ = -0.34 A | | 2.5 | 5 | Ω |
| | | $T_J$ = 125°C | | 4 | 10 | |
| | | $V_{GS}$ = -4.5 V, $I_D$ = -0.25 A | | 5.3 | 7.5 | |
| $I_{D(on)}$ | On-State Drain Current | $V_{GS}$ = -10 V, $V_{DS}$ = -10 V | -1 | | | A |
| $g_{FS}$ | Forward Transconductance | $V_{DS}$ = -10 V, $I_D$ = -0.34 A | | 250 | | mS |
| **DYNAMIC CHARACTERISTICS** | | | | | | |
| $C_{iss}$ | Input Capacitance | $V_{DS}$ = -25 V, $V_{GS}$ = 0 V, f = 1.0 MHz | | 40 | | pF |
| $C_{oss}$ | Output Capacitance | | | 13 | | pF |
| $C_{rss}$ | Reverse Transfer Capacitance | | | 4 | | pF |
| **SWITCHING CHARACTERISTICS** (Note 2) | | | | | | |
| $t_{D(on)}$ | Turn - On Delay Time | $V_{DD}$ = -25 V, $I_D$ = -0.25 A, $V_{GS}$ = -10 V, $R_{GEN}$ = 25 Ω | | 14 | 20 | nS |
| $t_r$ | Turn - On Rise Time | | | 6 | 20 | |
| $t_{D(off)}$ | Turn - Off Delay Time | | | 13 | 20 | |
| $t_f$ | Turn - Off Fall Time | | | 6 | 20 | |
| $Q_g$ | Total Gate Charge | $V_{DS}$ = -25 V, $I_D$ = -0.34 A, $V_{GS}$ = -10 V | | 1.3 | | nC |
| $Q_{gs}$ | Gate-Source Charge | | | 0.23 | | nC |
| $Q_{gd}$ | Gate-Drain Charge | | | 0.38 | | nC |

**ELECTRICAL CHARACTERISTICS** ($T_A$ = 25°C unless otherwise noted)

| Symbol | Parameter | Conditions | Min | Typ | Max | Units |
|---|---|---|---|---|---|---|
| **DRAIN-SOURCE DIODE CHARACTERISTICS** | | | | | | |
| $I_S$ | Maximum Continuous Source Current | | | | -0.34 | A |
| $I_{SM}$ | Maximum Pulse Source Current (Note 2) | | | | -1 | A |
| $V_{SD}$ | Drain-Source Diode Forward Voltage | $V_{GS}$ = 0 V, $I_S$ = -0.34 A (Note 2) | | -0.8 | -1.2 | V |

Notes:

1. $R_{\theta JA}$ is the sum of the junction-to-case and case-to-ambient thermal resistance where the case thermal reference is defined as the solder mounting surface of the drain pins. $R_{\theta JC}$ is guaranteed by design while $R_{\theta CA}$ is determined by the user's board design.

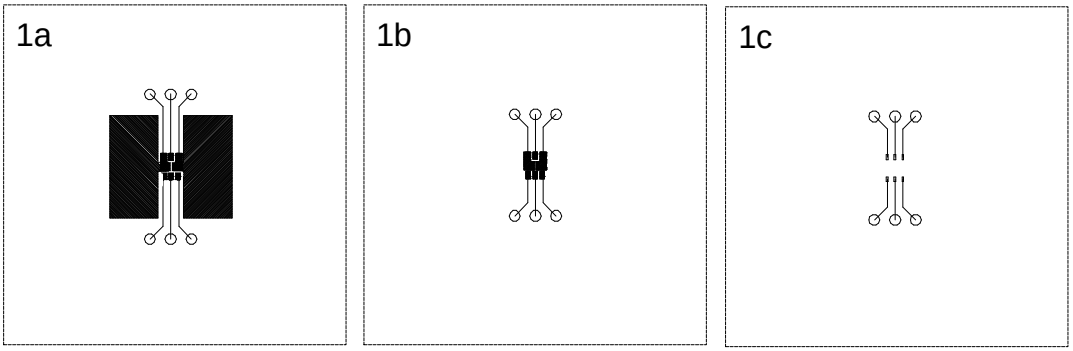
$$P_D(t) = \frac{T_J - T_A}{R_{\theta JA}(t)} = \frac{T_J - T_A}{R_{\theta JC} + R_{\theta CA}(t)} = I_D^2(t) \times R_{DS(ON)@T_J}$$

Typical $R_{\theta JA}$ for single device operation using the board layouts shown below on 4.5"x5" FR-4 PCB in a still air environment:

a. 130°C/W when mounted on a 0.125 $in^2$ pad of 2oz cpper.

b. 140°C/W when mounted on a 0.005 $in^2$ pad of 2oz cpper.

c. 180°C/W when mounted on a 0.0015 $in^2$ pad of 2oz cpper.




Scale 1 : 1 on letter size paper

2. Pulse Test: Pulse Width ≤ 300µs, Duty Cycle ≤ 2.0%.

## Typical Electrical Characteristics

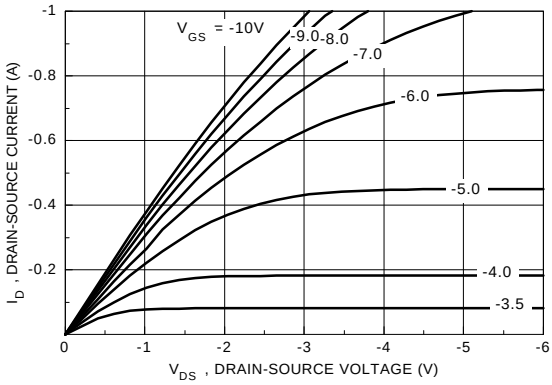



**Figure 1. On-Region Characteristics.**

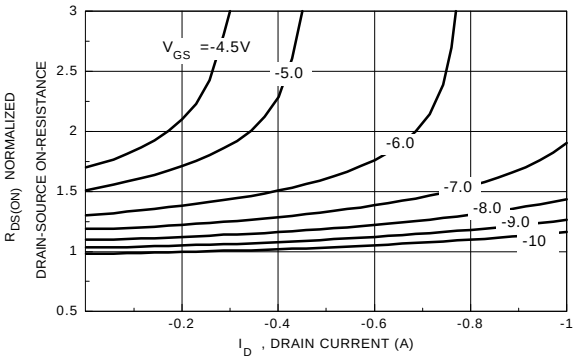



**Figure 2. On-Resistance Variation with Gate Voltage and Drain Current.**

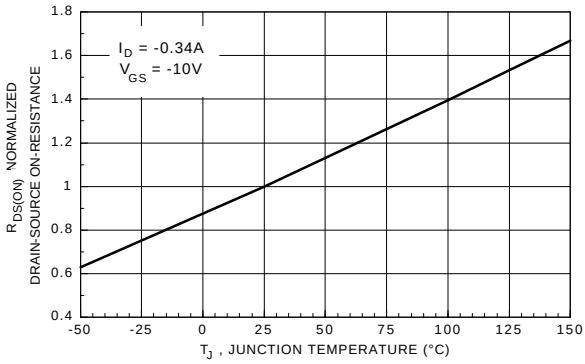



**Figure 3. On-Resistance Variation with Temperature.**

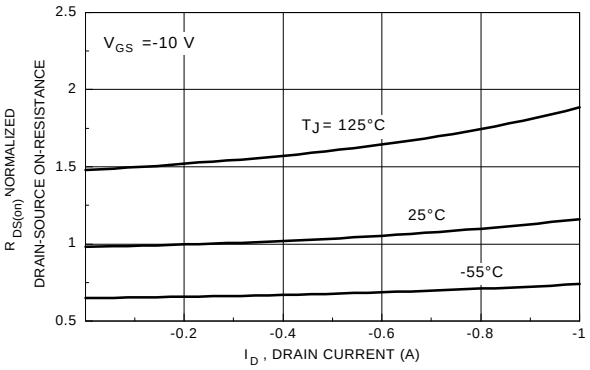



**Figure 4. On-Resistance Variation with Drain Current and Temperature.**

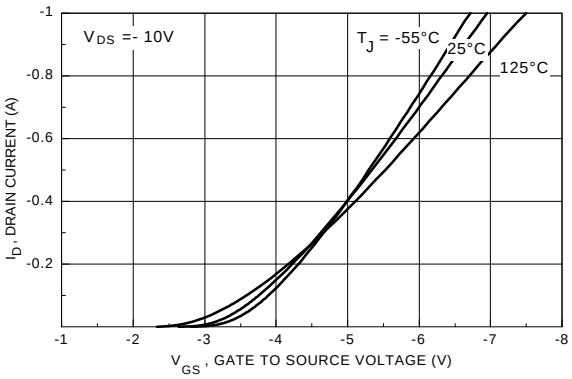



**Figure 5. Transfer Characteristics.**

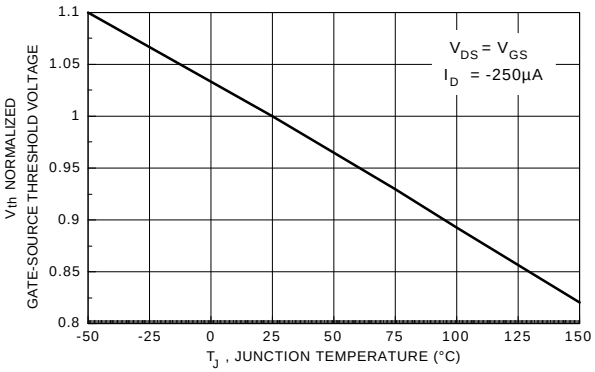



**Figure 6. Gate Threshold Variation with Temperature.**

## Typical Electrical Characteristics (continued)

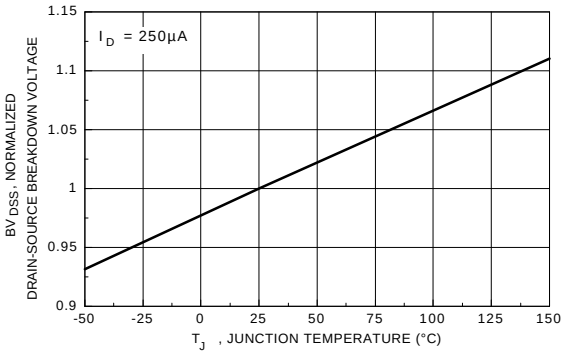



**Figure 7. Breakdown Voltage Variation with Temperature.**

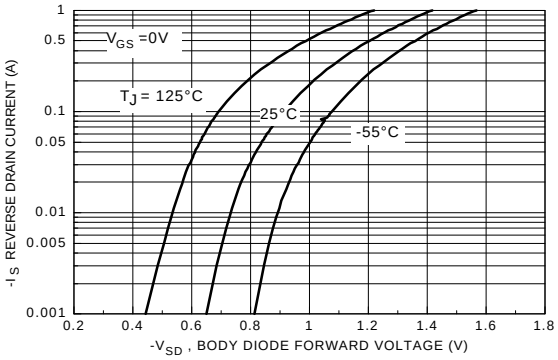



**Figure 8. Body Diode Forward Voltage Variation with Current and Temperature.**

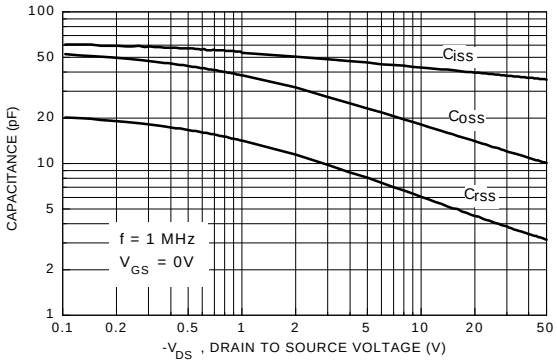



**Figure 9. Capacitance Characteristics.**

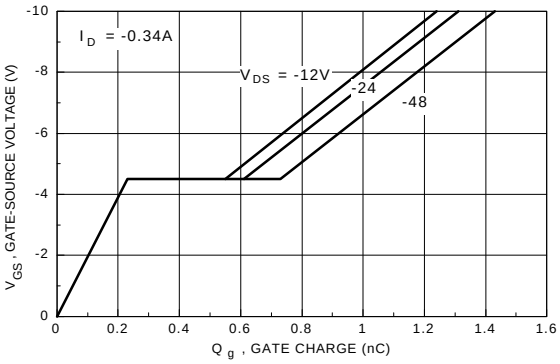



**Figure 10. Gate Charge Characteristics.**

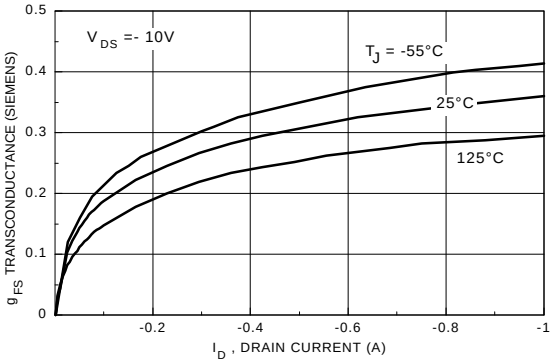



**Figure 11. Transconductance Variation with Drain Current and Temperature.**

## Typical Thermal Characteristics

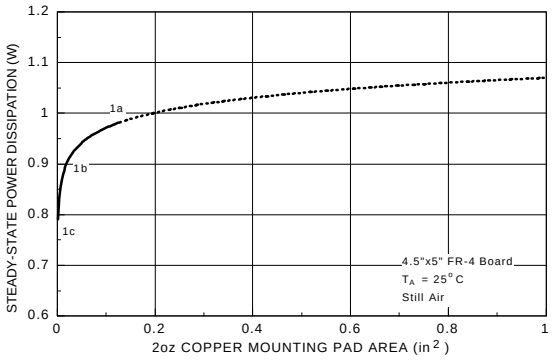



**Figure 12. SOT-6 Dual Package Maximum Steady-State Power Dissipation versus Copper Mounting Pad Area.**

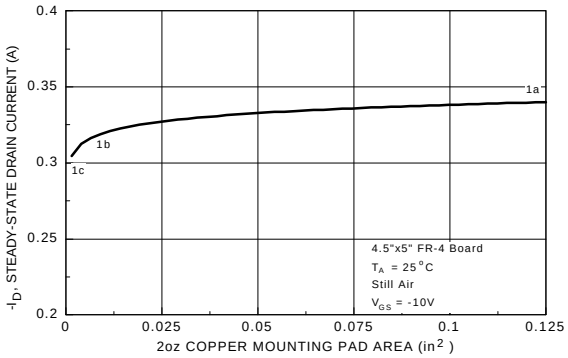



**Figure 13. Maximum Steady-State Drain Current versus Copper Mounting Pad Area.**

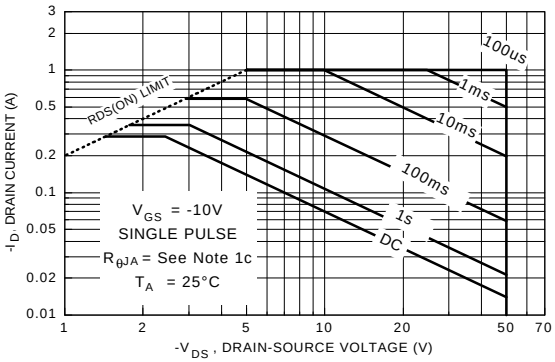



**Figure 14. Maximum Safe Operating Area**

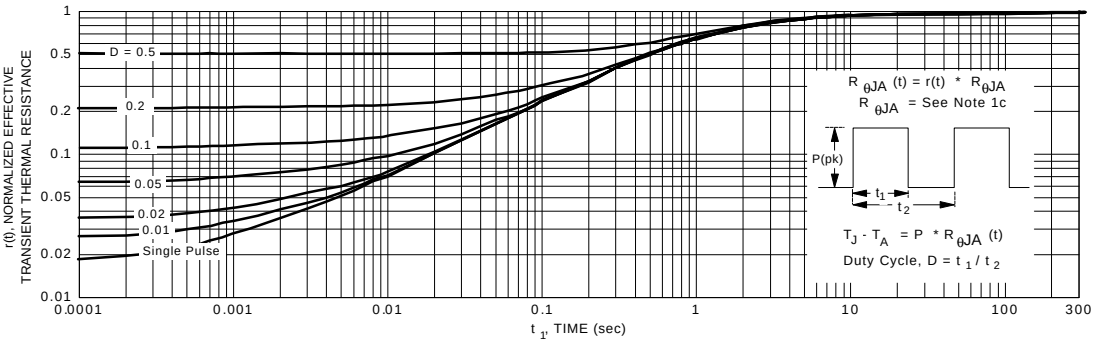



**Figure 15. Transient Thermal Response Curve.**

Note: Thermal characterization performed using the conditions described in note 1c. Transient thermal response will change depending on the circuit board design.

# SuperSOT™-6 Tape and Reel Data and Package Dimensions

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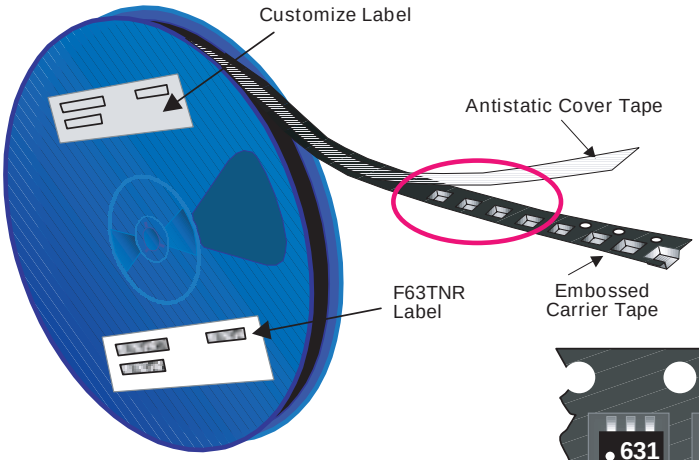
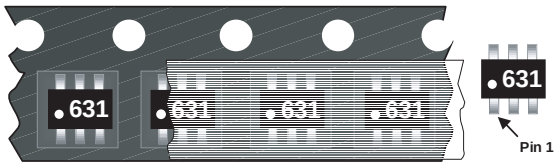
## SSOT-6 Packaging Configuration: Figure 1.0




**Packaging Description:**

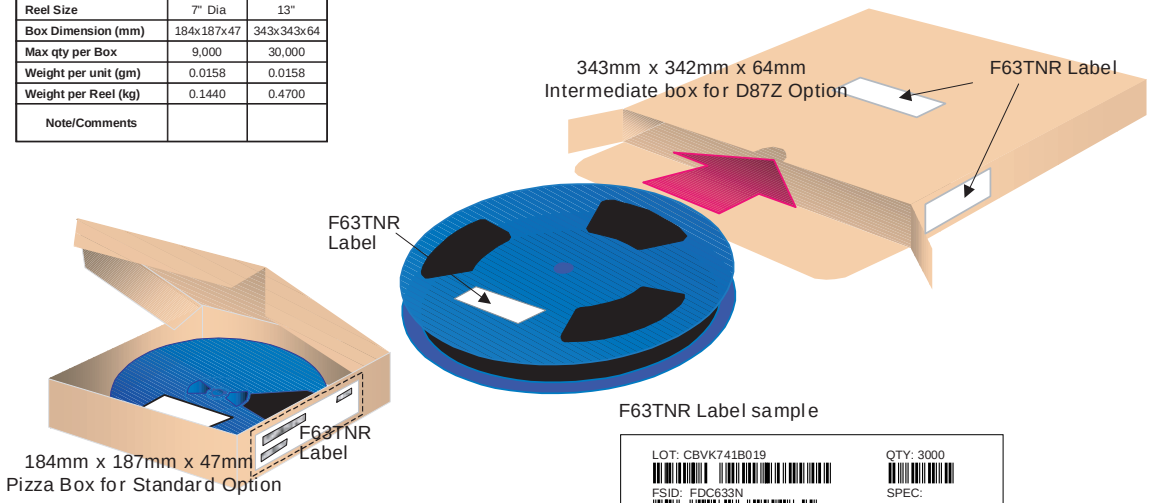
SSOT-6 parts are shipped in tape. The carrier tape is made from a dissipative (carbon filled) polycarbonate resin. The cover tape is a multilayer film (Heat Activated Adhesive in nature) primarily composed of polyester film, adhesive layer, sealant, and anti-static sprayed agent. These reeled parts in standard option are shipped with 3,000 units per 7" or 177cm diameter reel. The reels are dark blue in color and is made of polystyrene plastic (anti-static coated). Other option comes in 10,000 units per 13" or 330cm diameter reel. This and some other options are described in the Packaging Information table.

These full reels are individually barcode labeled and placed inside a pizza box (illustrated in figure 1.0) made of recyclable corrugated brown paper with a Fairchild logo printing. One pizza box contains three reels maximum. And these pizza boxes are placed inside a barcode labeled shipping box which comes in different sizes depending on the number of parts shipped.



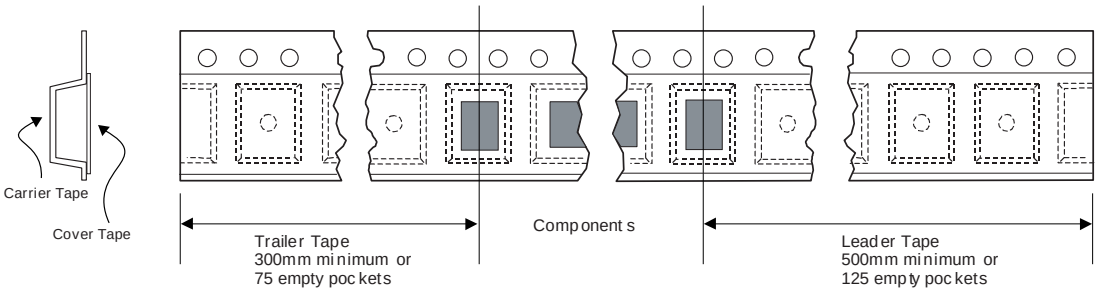

**SSOT-6 Unit Orientation**

| SSOT-6 Packaging Information | | |
|---|---|---|
| **Packaging Option** | Standard (no flow code) | D87Z |
| **Packaging type** | TNR | TNR |
| **Qty per Reel/Tube/Bag** | 3,000 | 10,000 |
| **Reel Size** | 7" Dia | 13" |
| **Box Dimension (mm)** | 184x187x47 | 343x343x64 |
| **Max qty per Box** | 9,000 | 30,000 |
| **Weight per unit (gm)** | 0.0158 | 0.0158 |
| **Weight per Reel (kg)** | 0.1440 | 0.4700 |
| **Note/Comments** | | |



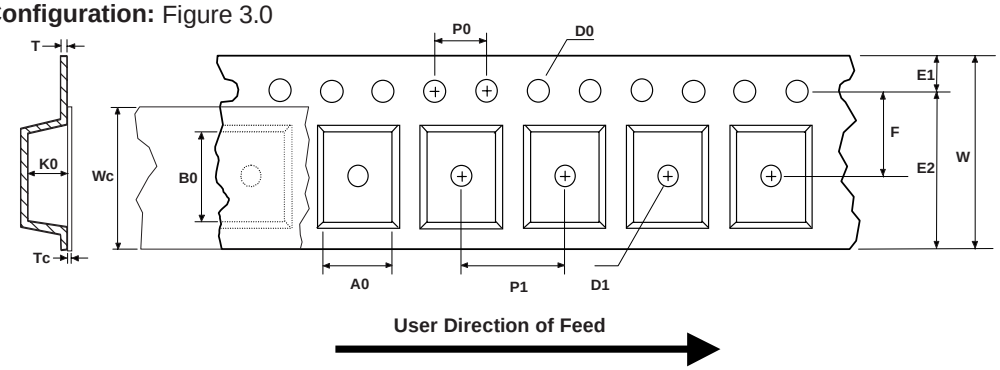

F63TNR Label sample

## SSOT-6 Tape Leader and Trailer Configuration: Figure 2.0



# SuperSOT™-6 Tape and Reel Data and Package Dimensions, continued

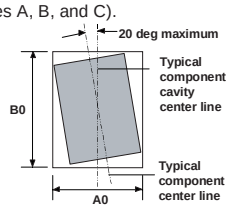
## SSOT-6 Embossed Carrier Tape
**Configuration:** Figure 3.0



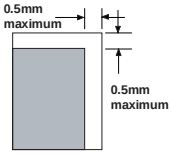

| Dimensions are in millimeter | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Pkg type** | **A0** | **B0** | **W** | **D0** | **D1** | **E1** | **E2** | **F** | **P1** | **P0** | **K0** | **T** | **Wc** | **Tc** |
| **SSOT-6** (8mm) | 3.23 +/-0.10 | 3.18 +/-0.10 | 8.0 +/-0.3 | 1.55 +/-0.05 | 1.125 +/-0.125 | 1.75 +/-0.10 | 6.25 min | 3.50 +/-0.05 | 4.0 +/-0.1 | 4.0 +/-0.1 | 1.37 +/-0.10 | 0.255 +/-0.150 | 5.2 +/-0.3 | 0.06 +/-0.02 |

Notes: A0, B0, and K0 dimensions are determined with respect to the EIA/Jedec RS-481 rotational and lateral movement requirements (see sketches A, B, and C).

Sketch A (Side or Front Sectional View)
**Component Rotation**

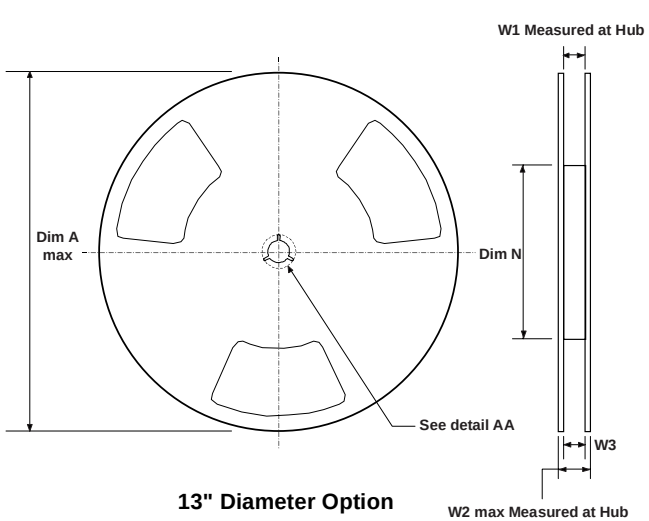



Sketch B (Top View)
**Component Rotation**

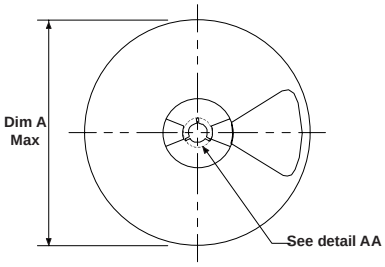



Sketch C (Top View)
**Component lateral movement**

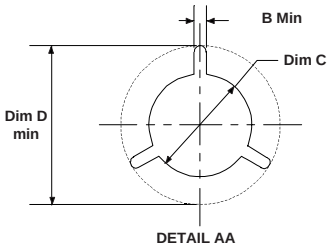
## SSOT-6 Reel Configuration: Figure 4.0




**13" Diameter Option**

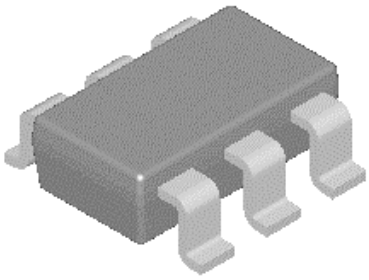


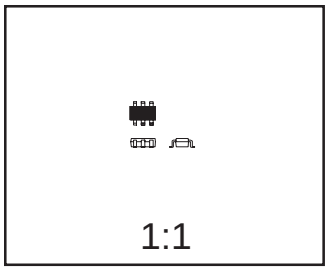

**7" Diameter Option**




| Dimensions are in inches and millimeters | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Tape Size** | **Reel Option** | **Dim A** | **Dim B** | **Dim C** | **Dim D** | **Dim N** | **Dim W1** | **Dim W2** | **Dim W3 (LSL-USL)** |
| 8mm | 7" Dia | 7.00<br>177.8 | 0.059<br>1.5 | 512 +0.020/-0.008<br>13 +0.5/-0.2 | 0.795<br>20.2 | 2.165<br>55 | 0.331 +0.059/-0.000<br>8.4 +1.5/0 | 0.567<br>14.4 | 0.311 – 0.429<br>7.9 – 10.9 |
| 8mm | 13" Dia | 13.00<br>330 | 0.059<br>1.5 | 512 +0.020/-0.008<br>13 +0.5/-0.2 | 0.795<br>20.2 | 4.00<br>100 | 0.331 +0.059/-0.000<br>8.4 +1.5/0 | 0.567<br>14.4 | 0.311 – 0.429<br>7.9 – 10.9 |

# SuperSOT -6 (FS PKG Code 31, 33)

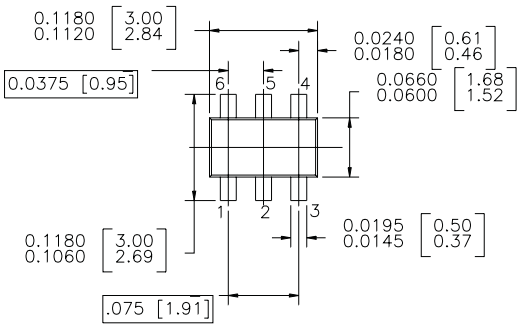




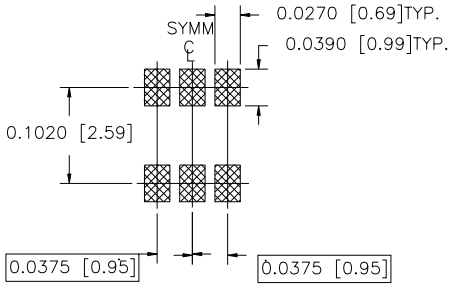

Scale 1:1 on letter size paper

Dimensions shown below are in:
inches [millimeters]

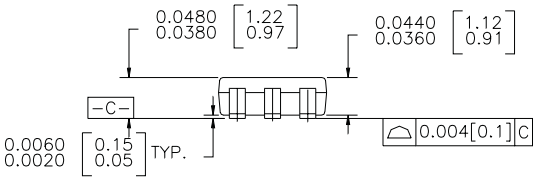
Part Weight per unit (gram): 0.0158

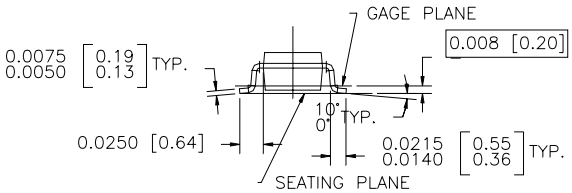



CONTROLLING DIMENSION IS INCH
VALUES IN [ ] ARE MILLIMETERS




LAND PATTERN RECOMMENDATION




NOTES : UNLESS OTHERWISE SPECIFIED

1.0 STANDARD LEAD FINISH : 150 MICROINCHES 93.81 MICROMETERS)
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2.0 NO JEDEC REGISTRATION AS OF JULY 1996

SUPER SOT 6 LEADS

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